STANDARD OPERATING PROCEDURE

SOP-03: ATS / ERS Calibration Compliance & Quality Assurance

Document ID:	SOP-03	**Effective Date:**	August 2026
Department:	Clinical Operations & QA	**Review Cycle:**	Annual
Approved By:	Robert E. Beaty, Founder	**Modality:**	Mobile Diagnostics

1.0 Purpose & Scope

1.1 Purpose: To establish standardized procedures for daily diagnostic equipment calibration, environmental parameter tracking, and test maneuver evaluation. This ensures that all clinical data generated by the PulmOne MiniBox+ strictly adheres to the acceptability and repeatability criteria published by the American Thoracic Society (ATS) and the European Respiratory Society (ERS).
1.2 Scope: This protocol dictates the daily startup requirements and real-time maneuver grading performed by the Supervisory Registered Respiratory Therapist (RRT) during all mobile PFT testing blocks.

2.0 Daily Volume Verification

Unlike legacy systems that require complex multi-gas calibrations, the MiniBox+ utilizes an advanced aerodynamic pneumotachograph. However, mechanical flow-volume accuracy must still be verified daily prior to patient testing.

1. **Frequency:** Volume verification must be performed once daily, prior to the first scheduled patient exam at the host clinic.
2. **Instrumentation:** The RRT will utilize a certified **3-Liter Calibration Syringe**. The syringe must be kept in the same ambient environment as the MiniBox+ for a minimum of 30 minutes prior to calibration to ensure thermal equilibrium.
3. **Execution:** The RRT will connect the 3L syringe to the device interface (using a fresh inline filter to replicate testing resistance) and inject/withdraw 3.0 Liters of air at varying flow rates (low, medium, and high flows) as guided by the MiniBox+ software.
4. **Tolerance Threshold:** The measured volume at all flow rates must fall within a strict $\pm$**3.0% tolerance limit** (2.91 L to 3.09 L). If the device fails, the flow sensor must be inspected for debris, reseated, and recalibrated. Testing may not commence until a passing calibration is logged.

3.0 BTPS Correction & Environmental Tracking

Spirometry and lung volume measurements must be corrected to Body Temperature and Pressure, Saturated (BTPS).

- **Internal Sensors:** The MiniBox+ contains internal sensors for ambient temperature, barometric pressure, and relative humidity.
- **Verification:** Upon daily boot-up, the RRT must verify that the internal environmental readings reflect the actual ambient conditions of the clinic exam room. If significant temperature shifts occurred during vehicle transit, the device must be allowed to acclimate to room temperature before BTPS correction values are established.

4.0 Clinical Acceptability & Repeatability Criteria

The RRT is responsible for coaching the patient to achieve the following ATS/ERS quality standards before concluding the diagnostic suite:

4.1 Spirometry (FVC & FEV1)

- **Acceptability:** Maneuvers must demonstrate a crisp, explosive start (Extrapolated Volume $<$ 5% of FVC or < 0.150 L). The maneuver must be free from artifacts, coughing during the first second, glottic closure, and early termination. Exhalation must last ≥ 6 seconds or achieve a definitive plateau on the volume-time curve (≤ 0.025 L change over 1 second).
- **Repeatability:** A minimum of three (3) acceptable maneuvers must be recorded. The difference between the two largest FVC values and the two largest FEV1 values must be ≤ 0.150 **Liters** (150 mL).

4.2 Gasless Lung Volumes (TLC & FRC)

- **Acceptability:** The patient must establish a stable end-expiratory tidal breathing baseline (FRC) prior to shutter closure. Panting maneuvers against the shutter must be executed at a consistent frequency (0.5 to 1.0 Hz) without air leaks around the mouthpiece.
- **Repeatability:** At least three (3) acceptable FRC plethysmographic measurements should be obtained, with the highest and lowest values agreeing within **5%.**

4.3 Single-Breath DLCO

- **Acceptability:** Rapid inspiration of test gas (≤ 4 seconds). Breath-hold duration must be strictly maintained between **8 to 12 seconds** without leaks or Valsalva/Müller maneuvers. Exhalation must be smooth to ensure an uncontaminated alveolar gas sample.
- **Repeatability:** A minimum of two (2) acceptable DLCO maneuvers must be captured. The values must agree within **2 mL/min/mmHg**.
- **Washout Delay:** The RRT must enforce a mandatory wait time of at least 4 minutes between DLCO attempts to allow for complete tracer gas washout from the lungs.

5.0 Automated Grading & Clinical Notation

The PulmOne MiniBox+ utilizes an automated quality-control algorithm (Grading A through F) based on ATS/ERS criteria.

- **Target Quality:** The RRT will strive to achieve ”Grade A” or ”Grade B” for all testing modalities.
- **Clinical Reality & Notation:** If a patient’s physical or cognitive limitations prevent them from achieving strict repeatability (e.g., severe COPD exhaustion, dementia), the RRT may conclude the test. The RRT **must** append clinical notes to the final report indicating the limitation (e.g., ”Patient unable to sustain 6-second exhalation due to severe dyspnea; best effort recorded”). This ensures the reviewing pulmonologist has complete context during the tele-overread process.

END OF STANDARD OPERATING PROCEDURE SOP-03